

8(a)	transition (emits) (one) photon with energy equal to the difference in energy between the two levels	B1
	frequency of radiation corresponds to energy of photon	B1
8(b)(i)	line to the left of the pair in Fig. 8.2, labelled A	B1
	larger gap between line A and the nearest of the pair in Fig. 8.2 than between the lines in the pair	B1
8(b)(ii)	line to the left of both the pair in Fig. 8.2 and line A, labelled B	B1
	larger gap between line B and line A than between line A and the nearest one of the pair in Fig. 8.2	B1
8(c)	$E = hf$	C1
	$E_3 = E_1 + h(f_A + f_B)$	A1